

Chapter 4

Static Analysis

Generally, analysis is broken down into i) stability and ii) system response to inputs. Both areas of study are vast enough to have their own dedicated topics. This brief discussion is intended to be an overview of the subject.

When transforming a system from the time-domain to the Laplace-domain the knowledge of time dependent characteristics is lost. However, two time-domain data points can be determined from the Laplace-domain transform: i) initial value of the output and ii) final value of the output. Additionally, there is a stability constraint on the evaluation of the output: the final value analysis is only valid when the output response is bounded as $t \rightarrow \infty$. Given a general transfer function

$$\frac{Y(s)}{R(s)} = \frac{b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0}$$

the output wrt to the input is written

$$Y(s) = \frac{b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0} \cdot R(s)$$

where b_i and a_i are real valued constants. If the system is stable and the input signal, $R(s)$ is known and bounded then the initial and final values of the output signal can be determined. The initial value of the output signal is $y(t = 0)$, while the final value of the output signal is $y(t \rightarrow \infty)$.

1. The **Initial Value Theorem (IVT)**

$$y(t = 0) = \lim_{s \rightarrow \infty} s \cdot Y(s) \quad (4.1)$$

2. The **Final Value Theorem (FVT)**

$$y(t \rightarrow \infty) = \lim_{s \rightarrow 0} s \cdot Y(s) \quad (4.2)$$

Note: in both theorems there is an additional s term that must be considered when solving for the output signal.

4.1 Initial Value Theorem (IVT)

Given a problem where the output signal is defined by

$$Y(s) = \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} \cdot R(s)$$

and the input signal is a setpoint, $R(s) = \frac{1}{s}$. Then, the initial value is

$$y(t=0) = \lim_{s \rightarrow \infty} s \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} \cdot \frac{1}{s}.$$

Simplify the expression and evaluate the limit

$$y(t=0) = \lim_{s \rightarrow \infty} \frac{4 + \frac{2}{s^2} + \frac{6}{s^3}}{1 + \frac{5}{s} + \frac{7}{s^2} + \frac{1}{s^3}} = \frac{4}{1} = 4$$

4.2 Final Value Theorem (FVT)

Given the same problem where the output signal is defined by

$$Y(s) = \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} \cdot R(s)$$

and the input signal is a step input, $R(s) = \frac{1}{s}$. Then, the final value is

$$y(t \rightarrow \infty) = \lim_{s \rightarrow 0^+} s \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} \cdot \frac{1}{s}$$

Now, either evaluate with L'Hôpital's rule or simplify the expression and evaluate the limit

$$y(t \rightarrow \infty) = \lim_{s \rightarrow 0^+} \frac{4s^3 + 2s + 6}{s^3 + 5s^2 + 7s + 1} = \frac{6}{1} = 6$$

The solution is only valid when the output $Y(s)$ is stable.

Recall: L'Hôpital's Rule

If, instead of algebraically manipulating the expression one prefers to use the more rigorous approach, here's the rule use

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

4.3 Zero — Pole Determination

We must use the open-loop model to determine the zeros and poles of a system

$$T(s) = \frac{B(s)}{A(s)}$$

Set $B(s) = 0$ and solve for the variable s , the solutions are the system zeros. Set $A(s) = 0$ and solve for the variable s , the solutions are the system poles. **Poles alone determine the system stability.** The system performance is determined by both the zeros and the poles.

Example

Given:

$$T(s) = \frac{s^2 + 14s}{s^4 + 5s^3 - 14s^2 - 140s - 200}$$

Find: The zeros and poles

The zeros:

$$\begin{aligned}s^2 + 14s &= s(s + 14) = 0 \\ s &= \{-14, 0\}\end{aligned}$$

The poles:

$$\begin{aligned}s^4 + 5s^3 - 14s^2 - 140s - 200 &= 0 \\ s &= \{-2, -4 \pm 2j, 5\}\end{aligned}$$

Note, the complex solutions for the poles. Zeros and poles may be real, complex, or imaginary.

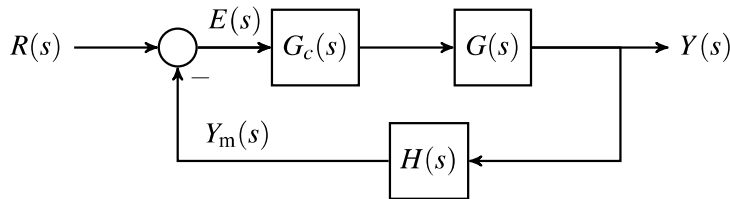
4.4 Steady State Error

Most controls problems involve the output signal tracking the input signal. Unimaginatively, we call these **tracking problems**. The typical implementation involves applying a desired or reference input to a system while designing a controller that modifies the natural behavior to the system so the output signal tracks the input signal after the transient (short term initial) response has settled.

The difference between the input and the output is the system error — at any point in time. When the system response has settled to a steady state, the error is then the steady state error, $e_{ss}(t \rightarrow \infty)$.

In the time-domain, $e(t) = r(t) - y(t)$. Where, $r(t)$ is the input **reference signal** and $y(t)$ is the ideal output signal. In the Laplace-domain, $E(s) = R(s) - Y(s)$. If the sensor feedback is poor, then the actual error as described the diagram $e(t) = r(t) - y_m(t)$ will differ from the ideal error in the above equation.

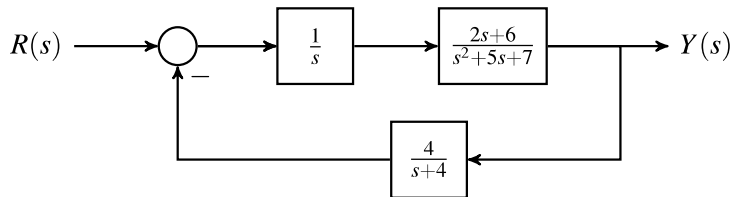
Steady state error analysis utilizes the negative feedback model, often with the controller explicitly defined in the forward path.



Predictably, there is more than one approach to solving these problems.

Option 1

Take a direct route by determining the transfer function $\frac{Y(s)}{R(s)}$ then using $E(s) = R(s) - Y(s)$.



where

$$Y(s) = \frac{\left(\frac{1}{s}\right)\left(\frac{2s+6}{s^2+5s+7}\right)}{1 + \left(\frac{1}{s}\right)\left(\frac{2s+6}{s^2+5s+7}\right)\left(\frac{4}{s+4}\right)} R(s)$$

which simplifies to

$$Y(s) = \frac{2s^2 + 14s + 24}{s^4 + 9s^3 + 27s^2 + 36s + 24} R(s)$$

So,

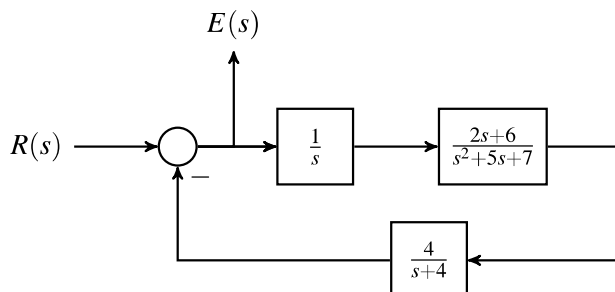
$$E(s) = R(s) - Y(s) = R(s) - \frac{2s^2 + 14s + 24}{s^4 + 9s^3 + 27s^2 + 36s + 24} R(s)$$

find the steady state response when $R(s) = \frac{1}{s}$

$$e(t \rightarrow \infty) = \lim_{s \rightarrow 0} s(R(s) - Y(s)) = \lim_{s \rightarrow 0} s\left(\frac{1}{s} - \frac{2s^2 + 14s + 24}{s^4 + 9s^3 + 27s^2 + 36s + 24} \frac{1}{s}\right) = 1 - \frac{24}{24} = 0$$

Option 2

Redefine the output to be the error signal and solve for the transfer function $\frac{E(s)}{R(s)}$.



Now,

$$E(s) = \frac{1}{1 + L(s)}$$

where the loop gain

$$L(s) = G_cGH(s) = \left(\frac{1}{s}\right)\left(\frac{2s+6}{s^2+5s+7}\right)\left(\frac{4}{s+4}\right)$$

Then

$$e(t \rightarrow \infty) = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} s\left(\frac{1}{1 + L(s)}\right)\frac{1}{s}$$

After simplification

$$e(t \rightarrow \infty) = \frac{1}{1 + L(0)}$$

where

$$L(0) = \lim_{s \rightarrow 0} \left(\frac{1}{s}\right)\left(\frac{2s+6}{s^2+5s+7}\right)\left(\frac{4}{s+4}\right) = \left(\frac{1}{0}\right)\left(\frac{6}{7}\right)\left(\frac{4}{4}\right) = \infty$$

which simplifies to

$$e(t \rightarrow \infty) = \frac{1}{1 + L(0)} = \frac{1}{\infty} = 0$$

Both approaches produce the correct solution, the first approach contains more algebraic manipulation than the second. Generally speaking, the second choice is the one with fewer possible pitfalls.

There is more on steady state error in Chapter 10.